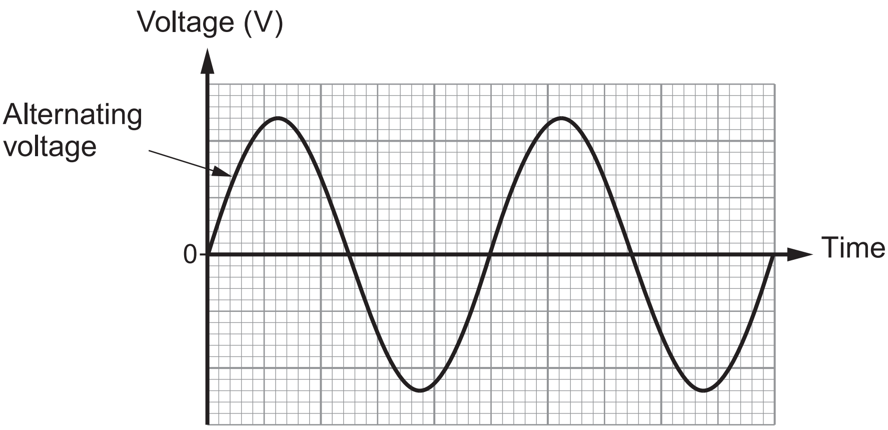
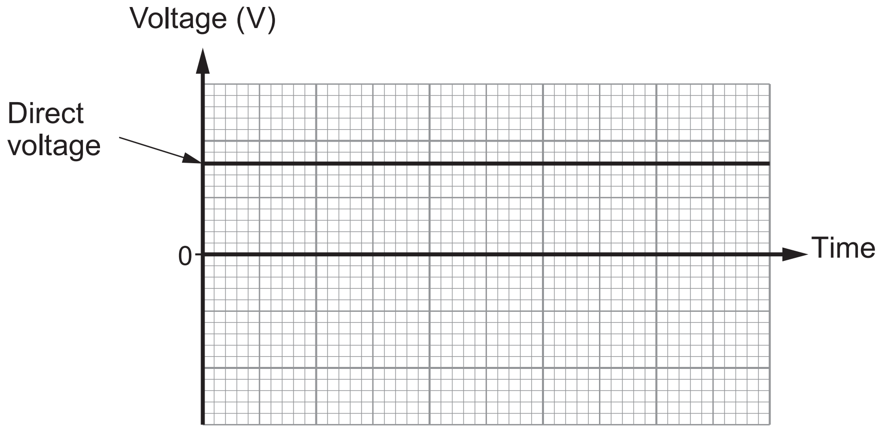
Examiner only

Answer **all** questions.

1. (a) The mains electricity supply to homes is at a voltage of 230 V a.c.
In contrast to this, batteries supply d.c.
These are each shown in the diagrams below.





An alternating voltage changes in size, but a direct voltage keeps the same size.
State **one** other difference between the two. [1]

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(b) Household circuits may be protected by a series of devices. These include:

- a miniature circuit breaker (mcb)
- a fuse in a plug
- a residual current circuit breaker (rccb).

The list below gives three situations in which one or another of these devices stop the power from being supplied.

1. An electric lawn mower cuts through the cable.
2. Too much current flows in a ring main because too many appliances are being used at the same time.
3. The live wire in a kettle touches its neutral wire.

(i) Select and explain one situation above that puts the **user** in danger of an electric shock **and** which one of the devices should be used to keep the user safe. [2]

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(ii) State **two** advantages of an mcb over a fuse in a household circuit. [2]

1. ....

2. ....

(c) Describe the function of the live and neutral wires in household circuits. [2]

Live wire: ....

....

Neutral wire: ....

....

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